RICA MOUELE

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LinkedIn | GitHub | Portfolio

PROFESSIONAL OBJECTIVE

I am looking for a **1-year apprenticeship starting in September 2026** in the fields of **Artificial Intelligence, Computer Vision and Data Science.**

Schedule:

-**First semester:** 3 days at school / 2 days in company.

-**Second semester: 90%** in company.

Driven by a strong desire to learn, I am committed to contributing my technical skills to innovative projects.

EDUCATION

2025-2027 | Université Paris Cité, Paris, France
Master in Computer Science, Vision and Machine Intelligence track

Key modules: Programming (Python, C++), Computer Vision (Image Analysis, 3D and Biomedical Imaging), Signal and Image Processing, Vision System Modelling, Deep Learning (Neural networks, Pattern recognition), Data Engineering (Data Science, Decision support), Applied Mathematics, Research Methodology..

2021-2024 | Institut Supérieur Informatique, Dakar, Sénégal
Bachelor in Computer Science, Software Engineering track

Key modules: Programming (Python, C/C++, Java), DevOps (CI/CD), Algorithms (data structures), Graph Theory, Applied Mathematics.

PROFESSIONAL EXPERIENCE

05/2025 - 09/2025 | IT Developer (Backend & Frontend)
FBNBank Senegal, Dakar - CDI

- Designed backend services (Java/.NET) coupled with graphical interfaces (React).
- Developed structured data processing pipelines following best development practices, ensuring security and elimination of human errors.
- Systematic production of logical flow diagrams, API documentation and maintenance guides.

Result: 95% automation of data entries, elimination of human errors, reduced operational lead times.

PROJETS UNIVERSITAIRES

2025 - 2026 | Coin Recognition and Classification (team + individual)

- Classical image analysis (team): Complete pipeline using Gaussian filters, Canny edge detection and Hough transform to segment coins, combined with texture and diameter analysis to automate classification and total amount calculation.
- Deep learning (individual): Development of a classifier based on AlexNet architecture to identify coin origin and value (Euros, Francs, etc.). Optimised feature extraction to outperform classical geometric methods.
- **Skills: Classification, comparative evaluation, false positive reduction.**

2025-2026 | Knowledge Distillation on CLIP Model (pair project)

- Implementation of the CLIP-KD paper (Yang et al.): distillation of semantic (image-text) knowledge from a large CLIP model (Teacher) to a compact architecture (Student) for Edge deployment.
- Adaptation to a noisy environment (underwater images), zero-shot evaluation, testing of different loss functions.
- Preserved image-text alignment, reduced computational cost.
- **Skills: Multimodal models, zero-shot evaluation, optimisation, fine-tuning.**

2025 - 2026 | Hybrid Information Retrieval Engine (team project)

- Design of a hybrid engine: Combining lexical (BM25) and semantic (Embeddings) search for multi-domain query processing.
- Development of a Reranking module and integration of frameworks (LangChain/LangGraph) to refine the relevance of results (Top-K). Optimization of Top-K accuracy and latency reduction on the dataset.
- **Skills: RAG, LLM, LangChain, performance evaluation, processing pipelines.**

TECHNICAL SKILLS

- **Languages:** Python (Data/IA), C/C++, SQL, JavaScript.
- **Artificial Intelligence & Deep Learning :** PyTorch, TensorFlow, Scikit-learn, CNN architectures (AlexNet, ResNet, DenseNet, ConvNext, GoogleLeNet, …), Transformers, image matching and retrieval methods, optimisation & distillation, RAG, LangChain, embeddings.
- **Image Processing & Computer Vision :** OpenCV, image dataset preparation and curation, segmentation, filtering, noise reduction, feature extraction, object detection (YOLO).
- **DevOps:** GitLab CI/CD, Git/GitHub, Docker.
- **Data Engineering :** PySpark, pipelines ETL, data manipulation.
- **R&D & Documentation :** Technology watch, implementation of state-of-the-art research, validation and statistical testing, technical documentation (notebooks, flow diagrams, commented code).

LANGUAGES

- French: Native.
- English: Upper-intermediate (B2).

INTERESTS

- Visual arts: Digital photography, graphic design
- Reading: Science-fiction novels, popular science
- Aeronautics & Aerospace (aviation technologies, space exploration)